CHEMICAL REACTIONS

Endothermic vs Exothermic — Fun Science

WHY DO SOME REACTIONS MAKE THINGS HOT OR COLD?

- Why do some reactions make things HOT and others make things COLD? Let's explore this amazing science mystery together!

WHAT IS A CHEMICAL REACTION?

- A chemical reaction happens when something changes into something new!
- Reactants → Products
- Examples: Roti cooking on tawa, burning diya, rusting iron gate
- More examples: Curd forming from milk, baking a cake
- Note: Melting ice and dissolving sugar are NOT chemical reactions

WHAT ARE EXOTHERMIC REACTIONS?

- "Exo = Exit. Heat exits and makes things HOT!" 🔥
- Memory Hook: Exo = Exit = Heat Exits = Gets HOT!
- Signal Words: gets hot, flame, burning, gives off heat

Exothermic Reactions Around Us

Warmth is everywhere in our daily life!
🔥 Burning coal in sigri – keeps us warm in winter
🔩 Rusting iron gate – releases heat slowly over time
🍲 Cooking dal on gas stove – heat makes yummy food!

WHAT ARE ENDOTHERMIC REACTIONS?

- "Endo = Enter. Heat enters and makes things COLD."
- Memory Hook: Endo = Enter = Heat Enters = Gets COLD 🧊
- Signal Words: "gets cold," "absorbs heat," "cools down," "needs heat"

Endothermic Reactions in Daily Life

Cooling happens all around us!

- Clay pot cooling water – Clay pot absorbs heat, keeping water cool
- Sweet soda + vinegar – Glass feels cold during reaction
- Photosynthesis – Plants absorb sunlight to make food

COMPARING EXOTHERMIC AND ENDOTHERMIC

Heat: Released (OUT) 🔥 vs Absorbed (IN) 🧊

Feel: Warm/Hot vs Cold/Cool

Examples: Burning diya vs Ice pack

Memory: Exo=Exit=Heat exits vs Endo=Enter=Heat in

LET'S SORT THESE REACTIONS!

Can you guess which is which?

1. Burning Diwali crackers → ?
2. Cold pack on sprained ankle → ?
3. Wood burning in Lohri bonfire → ?
4. Plants making food in sunlight → ?
5. Cooking dal on gas stove → ?
6. Baking soda + vinegar (cold glass) → ?

Why Do Reactions Make Things Hot or Cold?

- Reactions that give out heat make things HOT! 🔥
- Reactions that take in heat make things COLD! 🧊

EVERYDAY SCIENCE IS EVERYWHERE!

Chemical reactions happen all around us every day! From the kitchen to the garden, from festivals to the playground — science is everywhere you look. Keep your eyes open and notice the reactions happening in your own home and community!

QUICK RECAP

Chemical reaction = new substance formed

Exothermic = heat exits = feels HOT 🔥

Endothermic = heat enters = feels COLD 🧊

Examples: Burning diya (exo), Ice pack (endo)

SCIENCE SPY CHALLENGE!

Tonight, find ONE thing at home that gets HOT or COLD. Share your discoveries with the class tomorrow!

SEE YOU NEXT TIME!

Next time: We will do REAL experiments!
Thank you and goodbye!